



CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

Lecture No.- 12



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Recap of Previous Lecture



Topic

Connectivity in Graph Part-04



Topics to be Covered



Topic

Coloring in Graphs

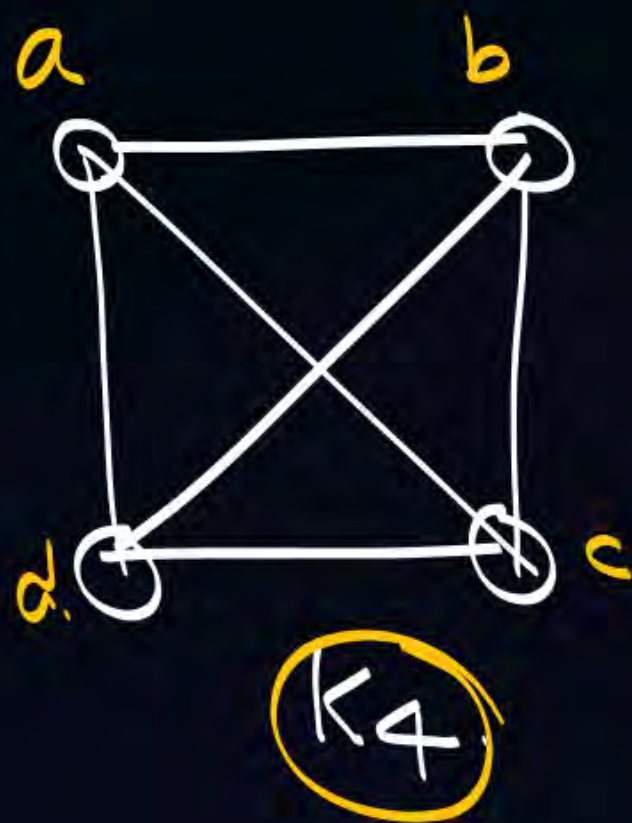


$$\frac{(n-1)!}{2}$$

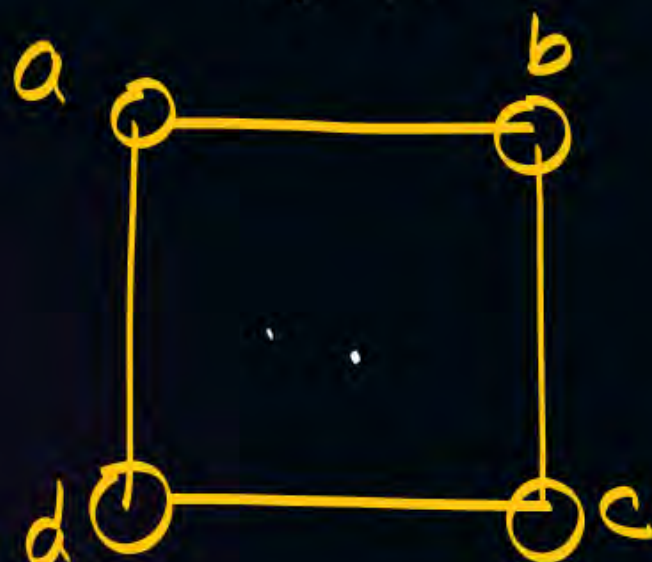
Total no of Hamiltonian cycle in K_n .

$$n=4.$$

$$\frac{3 \cdot 2 \cdot 1}{2} = \frac{3!}{2}$$



cycle 1



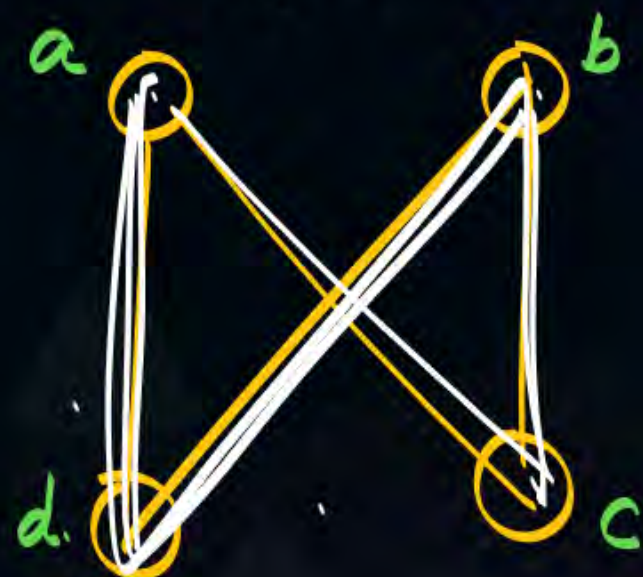
$\{a-b-c-d-a$
 $\{a-d-c-b-a$

cycle 2

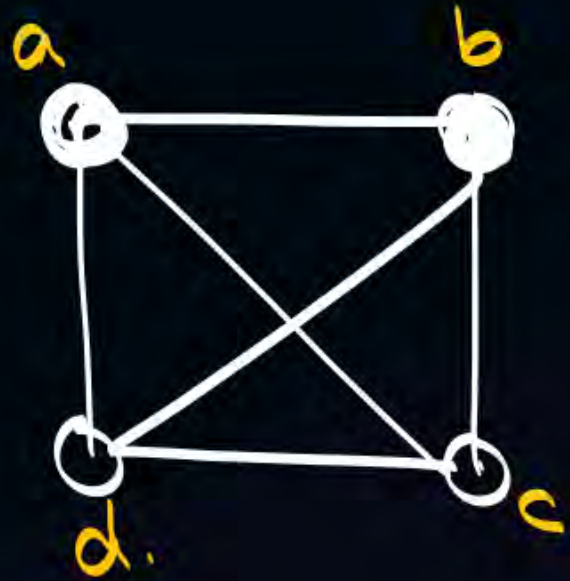


$\{a-b-d-c-a$
 $\{a-c-d-b-a$

cycle 3



$\{a-c-b-d-a$
 $\{a-d-b-c-a$



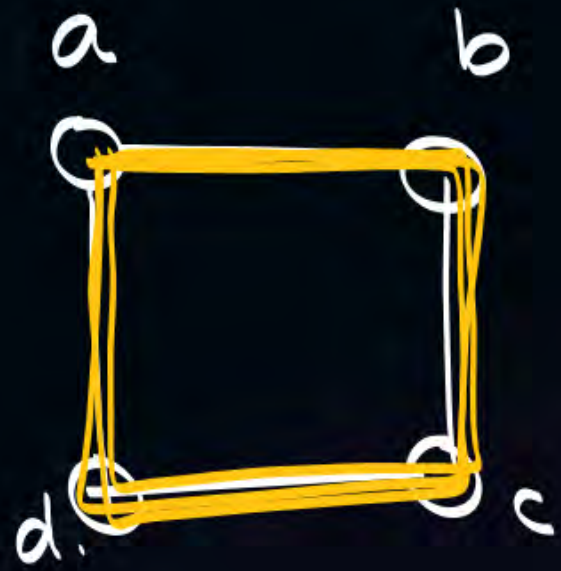
{ Start a $\rightarrow$ how many ways i can choose another vertex.

3 ways

after selecting 2nd vertex how many ways i can choose 3rd vertex

3. 2. 1.

2 ways



$$\begin{cases} a-b-c-d-a. \\ a-d-c-b-a. \end{cases}$$

no. of Hamiltonian cycle in $K_{n,n}$ ($n \geq 2$)

$$\frac{2 \cdot 1 \times 3 \cdot 2 \cdot 1}{(n-1)! \times n!} = \frac{n! (n-1)!}{2}$$

$K_{3,3}$



3 ways



2 ways

Left side

Right side

to choose any vertex.

2 ways

$\times$

1 way

3 ways

2 ways

$\times$ 1 way

$\Rightarrow$ if a Graph contains $\frac{(n-1)(n-2)}{2}$ no. of edges.

$$n = 5 \quad e = \frac{(5-1)(5-2)}{2}$$

Connected.

$$n = 5$$

$$e = 6$$



$$\begin{cases} n = 5 \\ e = 6 \end{cases} = \frac{4 \cdot 3}{2} = 6$$

Disconnected.

add one more $e = \frac{(n-k)(n-k+1)}{2}$
 $k = 2$



$$e = \frac{(n-2)(n-2+1)}{2}$$

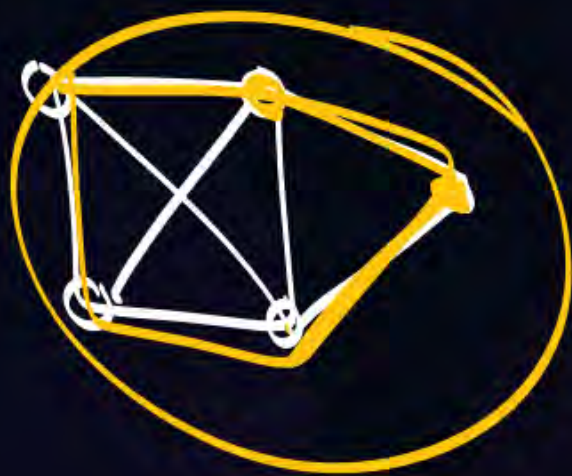
$$e = \frac{(n-2)(n-1)}{2}$$

- a) always connected.
- b) always disconnected.
- c) Sometimes Connected, Disc.
- d) not a.

Thm: if a Graph contains more than $\frac{(n-1)(n-2)}{2}$ no. of edges.
surely it will be connected.

Thm: if Graph contains $\frac{(n-1)(n-2)}{2} + 2$ no. of edges it contains Hamiltonian cycle.

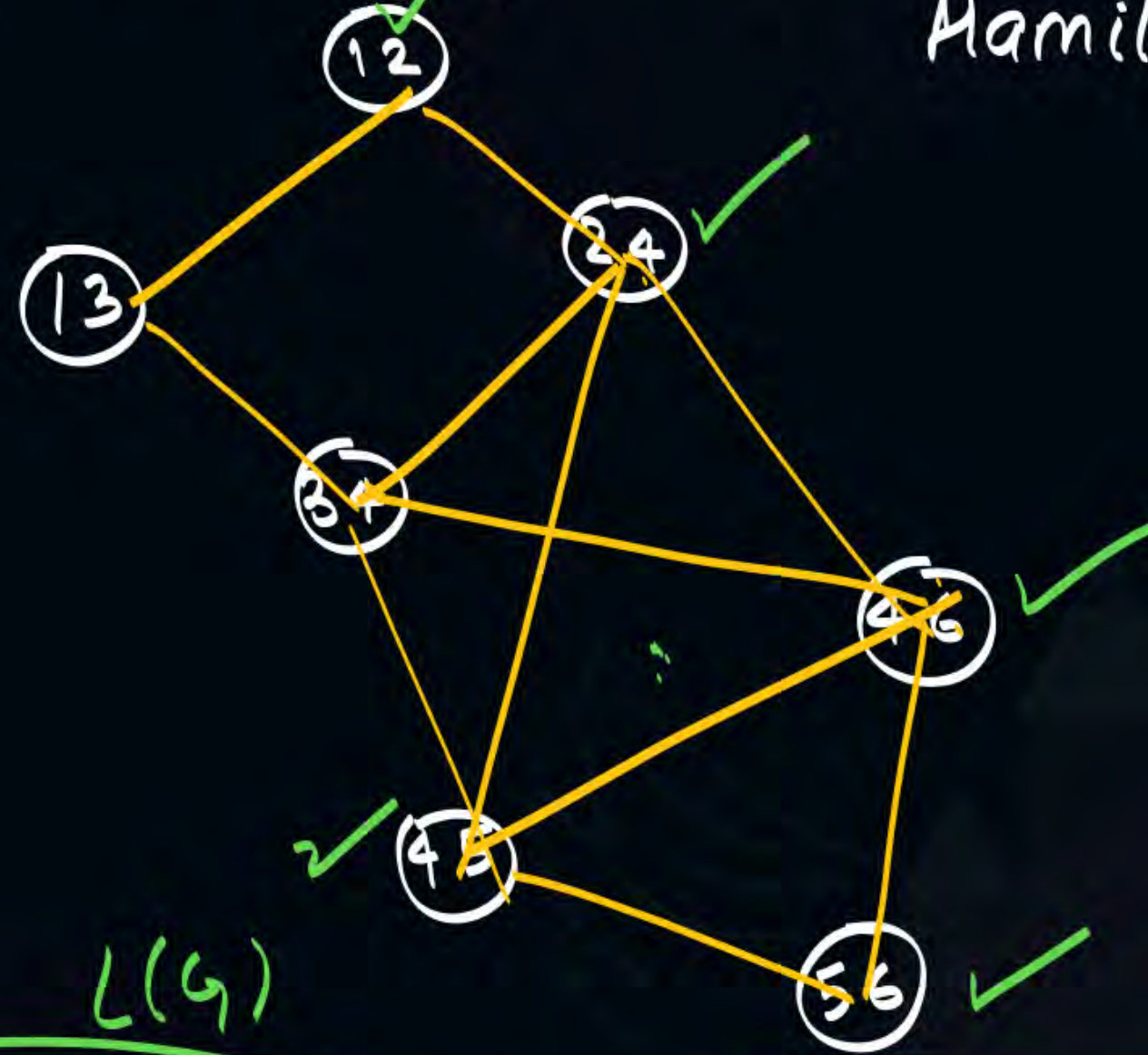
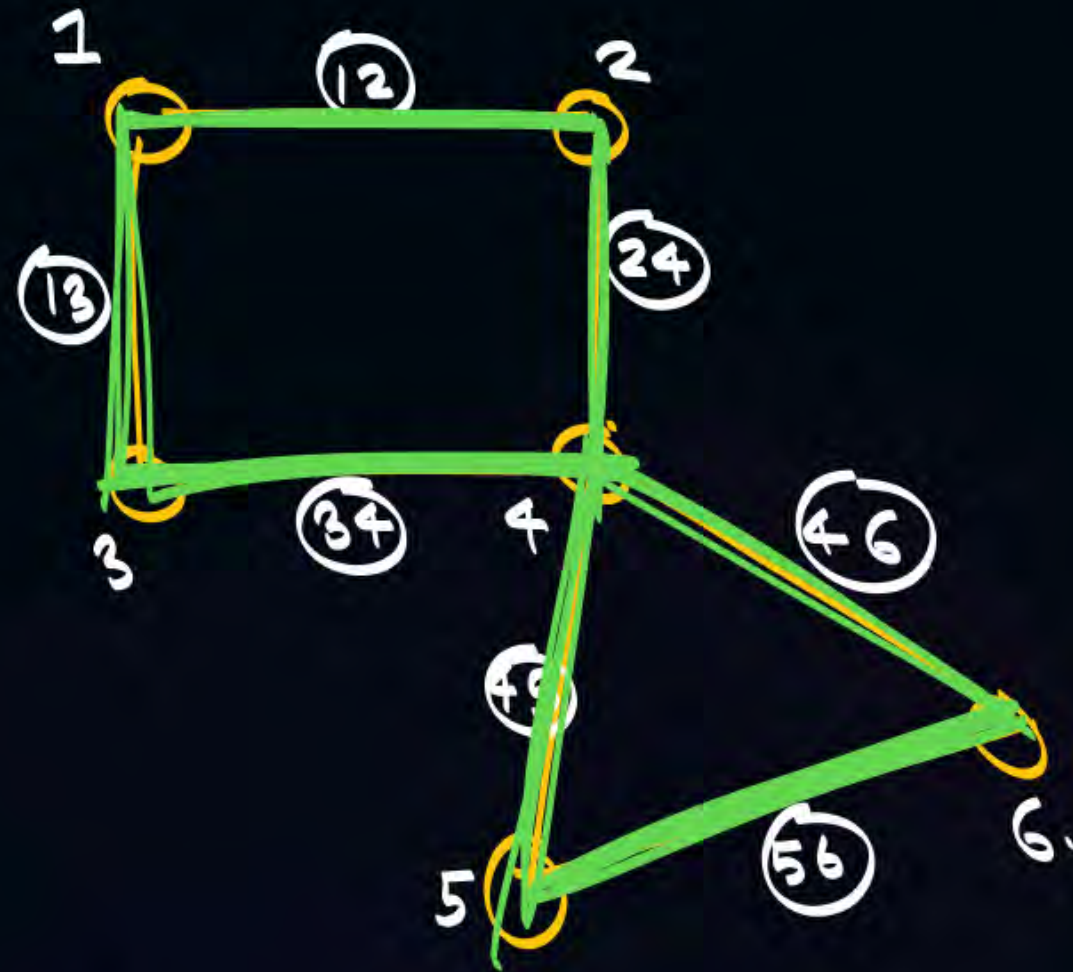
$n=5$ $e=6$.



$$\frac{(n-1)(n-2)}{2} + 1$$

H.P.

Line Graph of Euler Graph will always be Euler as well as Hamiltonian Graph.



Euler Graph.
all edges must be covered.

$L(G)$

vertices

edge disjoint Graphs.

2 graphs where, they don't have any common edges.

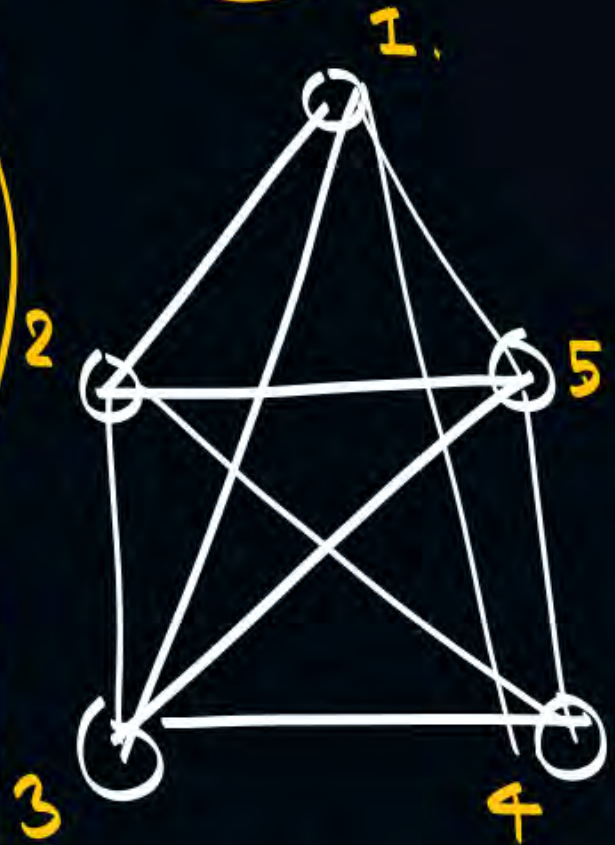
$$\frac{10 \text{ diff edges.}}{5}$$

= 2 diff-possibility

$$= \left\{ \frac{(n-1)}{2} \right\} \quad (n \text{ is odd})$$

$$\frac{n(n-1)}{2} \dots$$

10



5 edges.



edge disjoint
hamiltonian cycle.

5 students in a nursery play a game each day when they hold to form a cycle, how many days can they do this with no students holding hands with same player twice.

edge disjoint hamiltonian cycle



Hamiltonian cycle will requires n edges

Thm. no. of edge disjoint Hamiltonian cycle in K_n ($n \geq 3$, n is odd) will be $\frac{(n-1)}{2}$.

wait.

Coloring:

$\begin{cases} X(G) = K \\ K\text{-colorable Graph} \end{cases}$
 $X(G) = \text{properly coloring} + \text{min.}$

exp:

1 $\rightarrow$ all vertices with diff color. ~~X~~

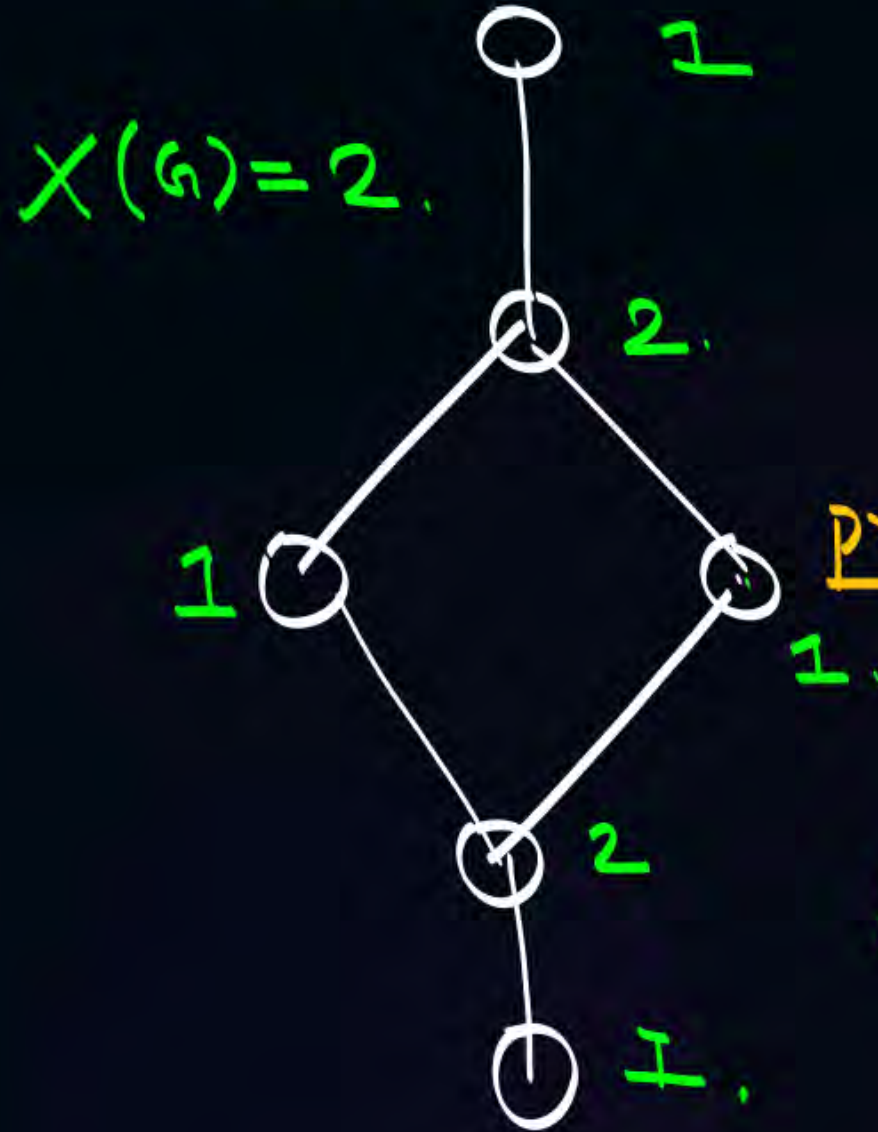
2 $\rightarrow$ min no. of colors. ~~X~~

properly coloring:

paint all the vertices with diff colors.
Such that adjacent should not have same color.

Chromatic no ($X(G)$)

paint all the vertices with min colors such that adjacent should not have same color.

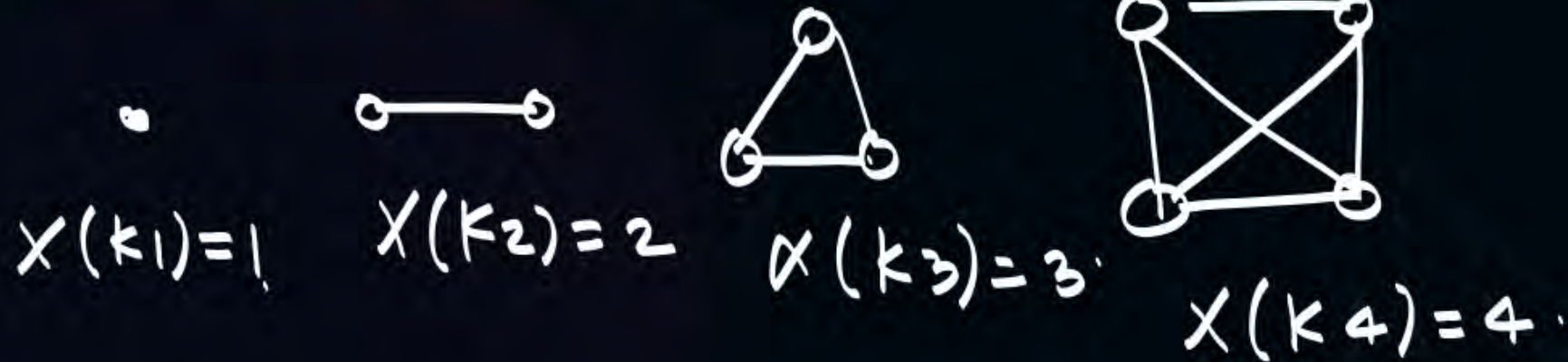


1) Isolated vertex:



$$X(G) = 1.$$

2) Complete Graph. (K_n)



$$X(K_1) = 1$$

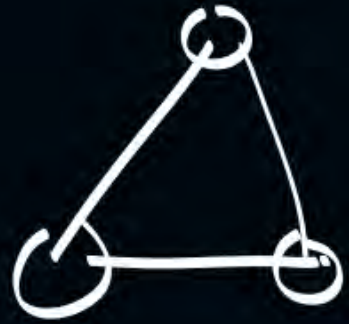
$$X(K_2) = 2$$

$$X(K_3) = 3$$

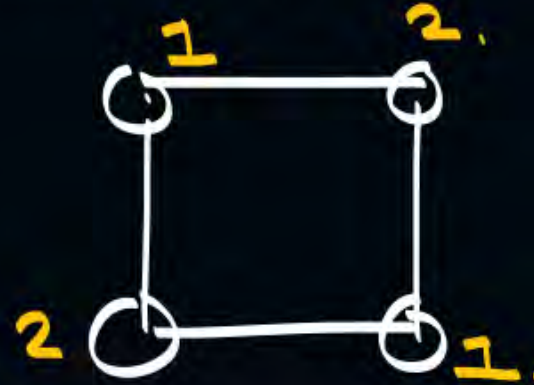
$$X(K_4) = 4$$

$$X(K_n) = n.$$

3) Cycle Graph:



$$\chi(C_3) = 3.$$



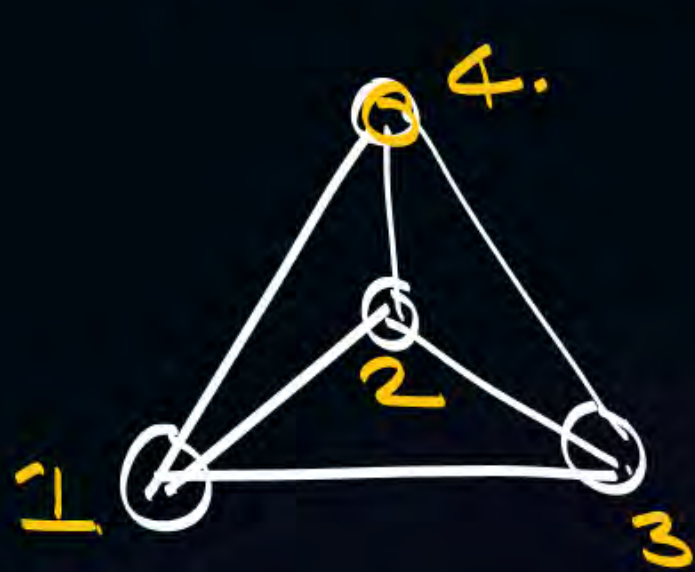
$$\chi(C_4) = 2.$$



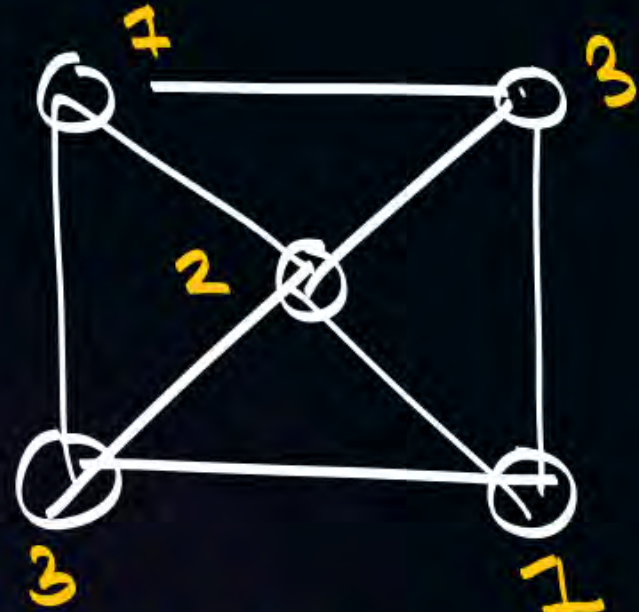
$$\chi(C_5) = 3.$$

$$\chi(C_n) = \begin{cases} 2 & \text{n is even} \\ 3 & \text{n is odd} \end{cases}$$

Wheel Graph:



$$\chi(W_4) = 4.$$



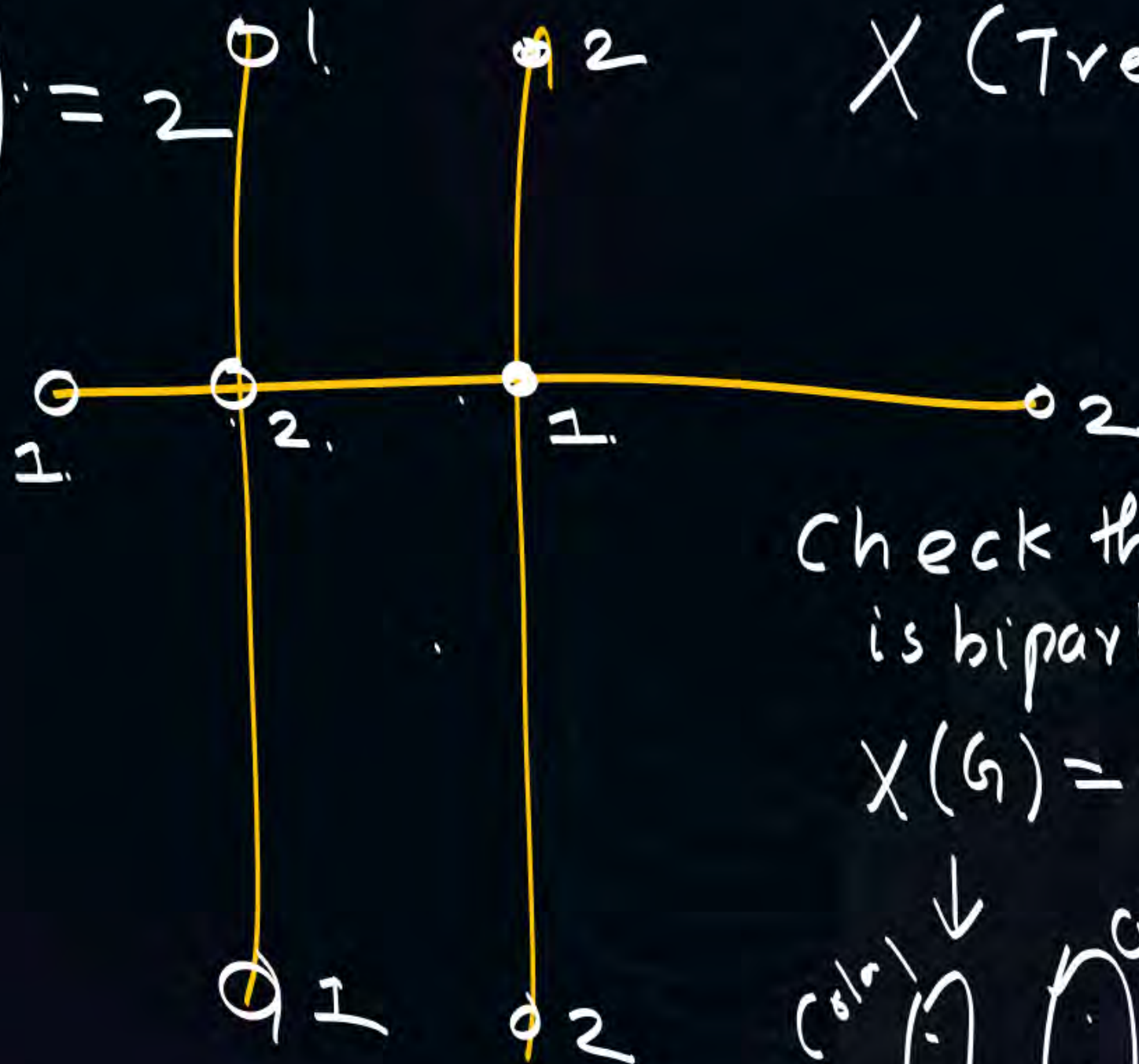
$$\chi(W_5) = 3$$

$$\chi(W_n) = 4. \quad n \text{ is even.}$$

$$\chi(W_n) = 3 \quad n \text{ is odd.}$$

$$\chi(\text{bipartite Graph}) = 2$$

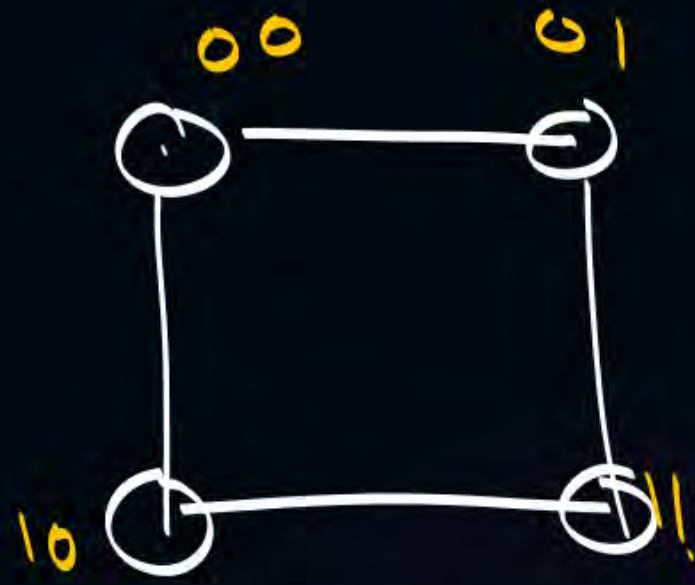
$$\chi(\text{Tree}) = 2$$



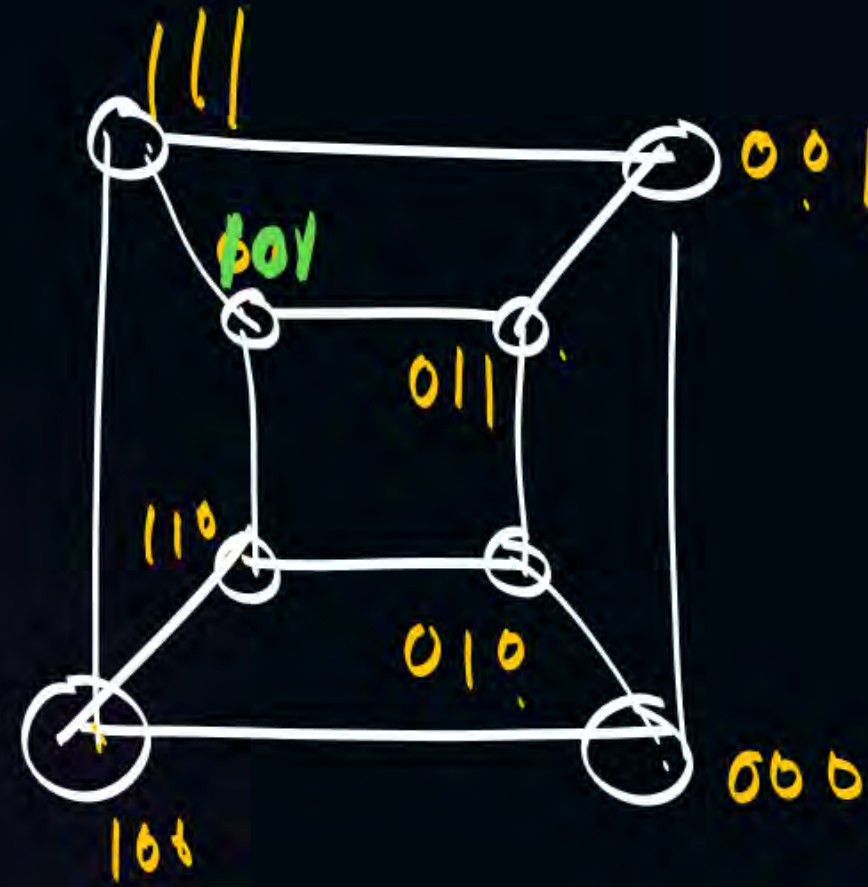
Check this
is bipartite or not.

$$\chi(G) = 2$$





$$\chi(G) = 2$$



This Graph. does not
contains odd length cycle.

↓
bipartite Graph.

↓
 $\chi(B.P.G) = 2.$



2 mins Summary

Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five

Coloring in Graphs

Trees $k=2$
forests

THANK - YOU

